

ABHIJEET RANJAN

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Education

Indian Institute of Information Technology and Management, Gwalior
Integrated B.Tech + M.Tech in Information Technology (Dual Degree); CGPA: 8.78/10

Gwalior, India
June 2025 – June 2030

Experience

Research Intern - Hardware Security and Side-Channel Analysis

Apr. 2026 – Jun. 2026

Indian Institute of Information Technology and Management, Gwalior

Gwalior, India

- Worked on hardware security experiments using the **ChipWhisperer** side-channel analysis platform to capture and inspect power traces from embedded cryptographic targets.
- Implemented **Differential Power Analysis** workflows covering trace acquisition, leakage modeling, hypothesis testing, and key-byte recovery on lab cryptographic workloads.
- Documented experimental setup, attack methodology, observations, and countermeasure considerations to support reproducible side-channel research.

Projects

LensBridge | *Tauri, React, Rust, WebRTC, DirectShow, Windows Local Services*

[GitHub](#)

- Built a local-first phone-to-Windows webcam bridge with **QR pairing, explicit desktop approval**, WebRTC streaming, live Tauri preview, trusted-device management, and local audit logs.
- Implemented Rust local signaling and a raw frame path into a Windows DirectShow camera through UnityCapture-compatible shared memory, with OBS fallback and quality profiles.

CTX | *Next.js, TypeScript, SQLite, Prisma, Browser Extension, MCP, GitHub APIs*

[GitHub](#)

- Built a local-first AI memory platform that converts chats, GitHub context, notes, and decisions into **searchable, reviewable, exportable, and injectable** context capsules.
- Shipped a Next.js dashboard, SQLite/Prisma persistence, browser extension workflows, MCP server tools, GitHub capture, secret redaction, snapshots, and backup/restore.

DeploySense | *TypeScript, Node.js, Next.js, Docker, Kubernetes, GitHub Actions, CLI*

[Live](#) | [GitHub](#)

- Engineered an open-source DevOps intelligence platform to scan Dockerfiles, Kubernetes manifests, GitHub Actions, Docker Compose, and deployment logs for production-risk issues.
- Implemented CLI/API/web dashboard workflows, **54+ scanner rules**, health scoring, SARIF/Markdown/JSON reports, Log Doctor failure chains, and conservative auto-fix flows.

GhostGate | *Linux, Bash, dnsmasq, iptables/nftables, Tor, WireGuard, Flask, VMware*

[GitHub](#)

- Built a VMware-based Ubuntu privacy router lab with **NAT, DNS resolution/logging**, tracker blocking, firewall baseline, traffic monitoring, Tor mode, and WireGuard routing paths.
- Created a GhostGate CLI, leak-test helpers, baseline scripts, and a read-only Flask dashboard for routes, DNS logs, firewall/NAT state, Tor/WireGuard status, and traffic counters.

Achievements and Leadership

Open Source Contributor: Selected for **GSSoC 2026**; contributing through issue-driven development, documentation, and collaborative engineering.

Google Cloud Certifications: Earned **18+ Skill Badges** across Generative AI, Cloud Run, Compute, Networking, Spanner, APIs, and Automation.

Technical Skills

Languages: C, C++, Java, Python, JavaScript/TypeScript, Bash, SQL, YAML

Frameworks: React, Next.js, Node.js, Express, Flask, Tauri, Prisma, Tailwind CSS, REST APIs

Developer Tools: Git, Docker, Docker Compose, Kubernetes, GitHub Actions, Linux, VMware, Vercel, Render, Nginx

Security/Networking: ChipWhisperer, Differential Power Analysis, Side-Channel Analysis, WebRTC, DirectShow, DNS, NAT, iptables/nftables, WireGuard, Tor, tcpdump, Nmap, Wireshark

Core Concepts: Operating Systems, OOP, DBMS, Computer Networks, Distributed Systems, API Design, CI/CD, Local-First Systems